GLDet: Real-Time SAR Ship Detector Based on Global Semantic Information Enhancement and Local Gradient Information Mining

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***Abstract*— Detecting ships in synthetic aperture radar (SAR) images is a challenging task due to various factors, such as the diverse distribution of ships and the intricate nature of SAR images. In recent years, deep learning has made excellent progress in the field of SAR interpretation. Models that focus on extracting global semantic information can effectively achieve balanced detection of multiscale SAR targets, but their computational complexity is relatively high. Models that focus on processing local information have redundant calculations and poor robustness, but are prone to mistaking the background information of SAR images for targets. To address the above issues, we propose a real-time SAR ship detector based on global semantic information enhancement and local gradient information mining. The lightweight feature extraction backbone based on linear computing is designed, with the network structure of global information augmentation encoder (GIAE), local gradient information miner (LGIM), and decoder, which can quickly perform feature extraction. GIAE enhances the expression of image content through the long-sequence modeling capability of the state-space model (SSM). LGIM uses gradient modules composed of depthwise separable convolutions to extract local information of image and utilizes directed self-attention (DSA) to mine channel context information. Global-local detector (GLDet) can complete object detection, rotated object detection, and instance segmentation tasks by transforming the detection head. Excellent performance has been achieved on the SAR ship instance segmentation datasets: SAR ship detection dataset (SSDD) and high-resolution SAR image dataset (HRSID), as well as the SAR rotated ship datasets: rotated ship detection dataset (RSDD)-SAR and SSDD+. Meanwhile, GLDet demonstrated excellent generalization performance in large-scale SAR images captured by GF-3 and Terra-SAR satellites.**

***Index Terms*— Instance segmentation, lightweight network, multitask learning, real-time detection, rotated object detection, ship detection, synthetic aperture radar (SAR).**

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I. Introduction

SYNTHETIC aperture radar (SAR) obtains images by actively emitting coherent electromagnetic waves to illuminate the surface and then receiving scattered echoes from surface targets [1]. It has all-time and all-weather image acquisition capabilities, making up for the shortcomings of optical imaging [2]. With the rapid development of SAR technology, multipolarized and multiband SAR data are becoming increasingly abundant, and the spatial resolution has reached the submeter level. At present, SAR plays a crucial role in multiple fields such as target reconnaissance and surveillance, strike indication, and natural disaster response [3].

Ship is the typical target of interest in SAR image interpretation. Timely and accurate disclosure of the number and distribution of ships can provide important information for battlefield intelligence reconnaissance and ocean management, which have important application value in military and civilian fields. Lincoln Laboratory proposed a three-level processing flow of detection–identification–classification to detect targets [4], [5].

This traditional method utilizes prior knowledge of the scene and target, and has good algorithmic interpretability. The most commonly used among them is the constant false alarm rate (CFAR) detector, which uses statistical tests to determine whether each pixel contains a target and can maintain a CFAR in cluttered backgrounds, thus gaining wide applications. However, they rely on manually designed features and complex parameter adjustments, which makes it difficult to establish reliable prediction models, and their robustness and scene generalization ability are poor [6], [7], [8].

At present, SAR interpretation tasks are moving toward finer granularity, and the demand for intelligent processing is becoming more urgent [9], [10]. Deep learning can automatically process complex image and learn the implicit internal relationships within them. Compared with traditional methods that utilize prior knowledge of targets and scenes, the introduction of deep learning technology significantly improves the performance of SAR target automatic target recognition [11], [12].

The existing deep learning networks mainly include convolutional neural networks (CNNs) and Transformer. CNN

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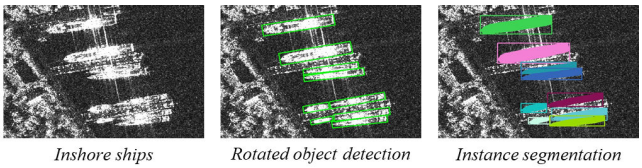


Fig. 1. SAR computer vision tasks.

is based on convolutional layers for feature extraction and focuses on local information (edges, textures, etc., of the target). Transformer directly captures the relationship between positions in the input image through self-attention mechanism, thereby obtaining global contextual information. The CNN model has a relatively simple structure, good parallelism, and computational efficiency, but poor robustness. Transformer has strong modeling capabilities, but it has high computational complexity and requires high hardware requirements. Therefore, it is necessary to design a more reasonable network architecture that combines the advantages of CNN in local feature extraction with the advantages of Transformer in global information modeling, in order to achieve a balance between inference delay and accuracy.

Instance segmentation and rotated object detection are important tasks in the field of SAR computer vision, as shown in Fig. 1. Instance segmentation aims to detect targets in the input image and assign category labels to each target pixel, which can effectively suppress background interference. Rotated object detection accurately identifies and locates specific targets by predicting the angle of the target box, which can effectively avoid the problem of overlapping prediction boxes. Computer vision algorithms based on deep learning networks are mainly divided into two-stage algorithms and one-stage algorithms. In the field of instance segmentation, two-stage algorithms include Mask region-based CNN (Mask R-CNN) series [13], [14], [15] and hybrid task cascade (HTC) [16]. One-stage algorithms include you only look at coefficient (YOLACT) [17], segmenting objects by locations (SOLO) series [18], [19], Mask2Former [20], RTMDet [21], ConvNeXt-V2 [22], and BoxInst [23]. In the field of rotated object detection, two-stage algorithms include rotated Faster-RCNN [24] and oriented R-CNN [25]. One-stage algorithms include rotated FCOS [26], R^3Det [27], S_2A-Net [28], and RTMDet [21]. The above algorithms are all universal detection frameworks, and the complexity and diversity of SAR images pose serious challenges to them.

The main limiting factors for high-precision SAR ship detection are differences in ship scale and strong scattering background interference. The following is a detailed analysis of the difficulties related to SAR ship detection.

1) In SAR images, different types of ships and different resolution images lead to differences in target scale, as shown in Fig. 2(a). Facing the problem of multiscale targets, using the fixed receptive field to extract feature cannot meet the detection requirements. As the network deepens, ship targets with weak scattering information or small pixels are prone to being missed. To ensure effective detection and recognition of targets, a more refined network structure design is needed.
2) The SAR image consists of bright pixels caused by strong scattering from various components of ships, and dark pixels caused by weak scattering from smooth areas such as the hull. The weak correlation between ship hull structures can lead to incomplete image features, and deep learning algorithms easily generate multiple prediction boxes for a single target [as shown in Fig. 2(b)]. The distribution of ships is random and the contour information is fuzzy, making it difficult to distinguish adjacent ships [as shown in Fig. 2(c)]. Therefore, accurately characterizing the scattering information of the target can enable the network to learn more discriminative intraclass semantic features, which is beneficial for improving the fine-grained recognition accuracy of the algorithm.
3) SAR and optical satellite sensors have fundamental differences in detection wavelength and imaging mechanism. In SAR images, buildings and metal facilities on the shore will generate strong echoes, presenting strong scattering areas like ships in the SAR images. These complex background clutter can cause visual confusion. Meanwhile, the imaging of ships anchored in the nearshore is relatively dark (with insignificant brightness information), which poses a significant challenge to the correct detection of targets, as shown in Fig. 2(d).

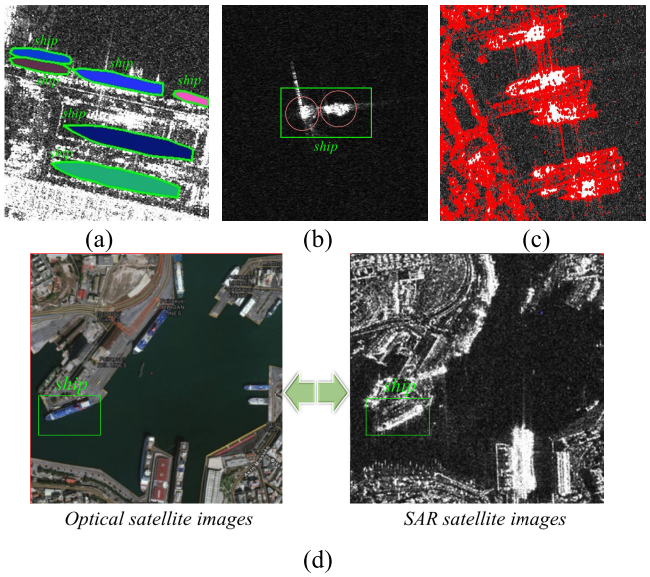


Fig. 2. Difficulties in SAR ship detection. (a) Multiscale ship image. (b) Discrete ship image. (c) Contour information of inshore ships. (d) Comparative images of optical satellites and SAR satellites.

At present, researchers have proposed many high-performance SAR ship instance segmentation and rotated box detection networks. Chen et al. [29] proposed a multitask learning instance segmentation method based on the segment anything model (SAM) [30] and demonstrated its effectiveness on the SAR ship detection dataset (SSDD) [31]. Yasir et al. [32] designed a CNN combined with a multihead self-attention (MHSA) mechanism, focusing on capturing the

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geometric features of ships, and validated its effectiveness on the OpenSARShip dataset [33]. Ju et al. [34] enhanced target features by introducing coordinate attention and proposed a dense regression heatmap loss function to improve detection performance, achieving excellent performance on the rotated SSDD+ [31] and rotated ship detection dataset (RSDD)-SAR [35].

Numerous scholars have designed detection networks based on CNN and Transformer by studying the complex background and speckle noise of SAR images, as well as the multiscale features of targets. However, existing SAR detection networks use many modules with high computational complexity, resulting in slow inference speed and inability to meet real-time requirements. Second, most networks have imbalanced extraction of global and local features from SAR images, resulting in weakened multiscale detection capabilities. At the same time, existing networks lack multitask learning capabilities, and most models can only meet a single task (such as object detection, semantic segmentation, and instance segmentation), resulting in poor generalization ability. The images obtained from SAR satellites have high resolution (HR), huge scale, complex scenes, and irregular target distribution. Most existing networks train and verify inshore and offshore scenes separately, lacking the detection ability for actual complex scenes.

Regarding the above issues and reviewing our previous work [36], [37], we have found that effective improvement methods for deep learning in the field of SAR detection are as follows.

1) The shallow features extracted by the backbone network contain more contour feature information, which is suitable for detecting small targets. Deep features contain semantic features of the image, which are suitable for distinguishing from background interference and detecting large targets.
2) Feature pyramid network (FPN) [38] is a simple and effective feature fusion method that can significantly improve the ability to extract multiscale features, making the model more robust when dealing with ship targets of different scales.
3) The self-attention mechanism can capture global contextual information in images and has significant advantages in dealing with complex backgrounds and speckle noise, demonstrating good application prospects in SAR image detection.
4) Multitask learning can improve the generalization ability and segmentation accuracy of models by jointly training multiple related tasks. By refining the detection and segmentation results of the model, it can better adapt to complex SAR image scenes.

In this article, we propose a real-time SAR ship detector based on global semantic information enhancement and local gradient information mining [global-local detector (GLDet)]. The detector continues the style of the you only look once (YOLO) series detectors and consists of a feature extraction backbone, FPN, and detection head.

In the task of extracting global information from images, both Mamba [55] and Transformer [54] are capable of efficiently capturing global dependencies. However, Mamba demonstrates superior performance due to its unique operational paradigm. GLDet adopts Mamba as the image encoder, leveraging its state-space-based architecture with linear computational complexity to address the specific challenges of SAR ship detection. Compared to Transformer, Mamba offers three key advantages.

1) *Linear Computational Complexity:* Unlike Transformer's quadratic self-attention mechanism, Mamba achieves linear scaling with sequence length through its selective state-space formulation. This is critical for processing HR SAR images.
2) *Dynamic Feature Selection:* Mamba's input-dependent selection mechanism adaptively focuses on ship-related features (e.g., bright scattering points and hull contours) while suppressing background interference from coastal structures. This mimics the CFAR principle from traditional SAR detectors, enabling better adaptation to diverse inshore/offshore scenarios compared to the Transformer's static attention patterns.
3) *Long-Range ContextModeling*: CNN is difficult to extract global dependence information due to its limited receiving domain, and Transformers face computational bottlenecks. However, Mamba efficiently models interactions between ships and complex backgrounds [e.g., Fig. 2(d)] through its hidden state propagation mechanism. This is particularly advantageous for resolving ambiguities between ships and onshore structures with similar scattering characteristics.

While Mamba excels at global semantic modeling, effective SAR ship detection also requires sophisticated local feature extraction for fine-grained structural pattern recognition and suppression of background artifacts. To this end, GLDet introduces a local gradient information miner (LGIM) framework comprising two synergistic components.

1) *Multiscale Gradient Fusion:* The gradient information mining (GIM) module employs a drip-shaped RepVGG-Block cascade to generate gradient-aware feature maps. GIM explicitly quantifies each pixel's contribution to loss dynamics through gradient magnitude reflection. This biologically inspired mechanism mimics human visual saliency. Large gradients highlight features critical for ship discrimination—such as deck edges and radar reflector corners—while suppressing low-contribution background pixels.
2) *Gradient-Guided Channel Pruning:* GIM's multibranch outputs feed into our directed self-attention (DSA) module, which performs channel-level feature recalibration. Channels with weak feature information are attenuated using learnable decay factors.

By integrating Mamba with CNN components, our architecture balances local feature extraction and global context modeling, addressing SAR-specific challenges including multiscale targets, background clutter, and real-time processing requirements. Our main contributions are as follows.

1) GLDet innovatively uses linear operation modules to balance the extraction of global and local informa-

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tion, highlighting target information and suppressing background interference, enabling the network to have extremely fast detection and inference speeds.
2) GLDet can complete different tasks by transforming the detection head and can achieve multitask learning.
3) Compared with existing excellent models, GLDet exhibits excellent performance in object detection, rotated object detection, and instance segmentation on multiple SAR ship datasets.
4) In large-scale SAR images captured by GF-3 and TerraSAR satellites, GLDet exhibits excellent generalization performance and can achieve accurate detection of ships.

II. Related Works

A. Instance Segmentation

Instance segmentation is a fundamental task for understanding image and video scenes. With the rapid development of deep CNNs, significant progress has been made in instance segmentation techniques, mainly including two-stage instance segmentation methods and one-stage instance segmentation methods.

The two-stage instance segmentation method first uses a detector to generate candidate regions, and then segments the candidate regions and generates pixel-level masks for each instance. Mask R-CNN [13] extends Faster R-CNN by adding a mask prediction branch to segment targets in candidate boxes. This algorithm efficiently utilizes the detection and segmentation stages, greatly improving segmentation accuracy. Subsequent algorithms such as Mask scoring R-CNN [15], Cascade R-CNN [14], and HTC [16] are improvements based on the Mask R-CNN framework. Mask scoring R-CNN alleviates the bias between mask quality and scoring, Cascade R-CNN fully utilizes the interrelationships between detection and segmentation to achieve multitasking learning, HTC interweaves detection and segmentation features for joint multistage processing to achieve more accurate predictions.

There are various forms of one-stage instance segmentation methods. YOLACT [17] predicts a set of prototype masks and mask coefficients for each instance, and combines them through matrix multiplication. SOLO [18] defines the instance category as the location and size of the target, and divides the instance segmentation task into category prediction and generating instance masks. Mask2former [20] uses a transformer structure to extract global features and utilizes deformable attention to capture the target contour. ConvNeXt-V2 [22] uses a fully convolutional mask self-encoder framework and a global response normalization (GRN) layer to improve multiscale detection capabilities. BoxInst [23] uses a fully convolutional instance segmentation framework to complete instance segmentation with only target box calibration. RTMDet [21] is based on the YOLO architecture, which increases the receptive field by introducing larger convolutional kernels and achieves a balance between computational efficiency and accuracy by optimizing training strategies. One-stage methods can simultaneously locate, classify, and segment objects, with better generalization performance.

In the field of SAR instance segmentation, scholars mainly design feature extraction modules or attention modules based on the above models, focusing on the scale features, scattering features, and background interference of SAR images. Based on a two-stage algorithm, Zhang and Zhang [43] embedded a full-level context module in the network to obtain more prominent feature maps, thereby reducing background interference. Wei et al. [41] introduced super-resolution (SR) denoising technology to capture image-level regions and texture information, and used HR technology to enhance Mask region of interest (RoI) features, achieving certain performance improvements. Based on a single-stage algorithm, Gao et al. [44] proposed a center point prediction algorithm to solve the difficult detection challenge of densely arranged targets, effectively improving the algorithm's ship contour segmentation performance. Yasir et al. [66] improved the backbone network structure by combining effective long-range aggregation network (ELAN) Net backbone network and FPN, which enhanced the network's predictive ability for multiscale targets. The above algorithms have achieved excellent detection results, but there is a certain degree of computational redundancy and mutual influence between network modules, resulting in a poor balance between detection accuracy and speed.

B. Rotated Object Detection

Rotated object detection is an important branch of object detection aimed at detecting target objects that appear at any angle in the image.

Rotation region proposal network (RRPN) [53] introduces rotated FCOS candidate boxes based on the region proposal network (RPN), achieving object detection in any direction for the first time. Many subsequent methods directly regress the five parameters of the rotated box (center coordinate, width, height, and rotated angle), such as oriented R-CNN [25], rotated FCOS [26], R^3Det [27], and S_2A-Net [28]. Based on inheriting the advantages of traditional R-CNN series methods, oriented R-CNN achieves fast and accurate detection of rotating targets by introducing directed candidate box generation network and directed object detection head. Rotated FCOS is a one-stage anchor-free rotating object detector consisting only of convolutional layers, which uses an elliptical center sampling method to define appropriate sampling areas for oriented bounding boxes (OBBs), making the network architecture more lightweight. R^3Det proposes a deducible approximate SkewIoU loss on the one-stage detector. S_2A-Net introduces a feature alignment module and a directed detection module on the one-stage detector to improve localization accuracy.

At present, one-stage detectors are favored by scholars due to their lightweight network structure and fast detection speed. One-stage detector consists of a backbone network, a feature fusion network, and a detection head. Wan et al. [47] designed an adaptive boundary enhancement module in the network to align multilevel fine-grained semantic information and suppress background interference. Wang et al. [48] focused on the acquisition of global information, designed the context semantic information fusion module and scattering point information learning module, and realized the accurate detection of multiscale ships. Yue et al. [49] designed a new network to realize directional ship detection through the

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annotated data in the horizontal bounding box (HBB). The above model has achieved good results, but there may be redundancy in the reasoning process of the network, resulting in a long detection time.

C. Deep Learning Networks

CNN has good performance in image processing, capable of processing complex image features. It performs well in processing local features, but performs weakly in processing global information. Transformer [54] has shown good performance in the field of natural language processing (NLP) through self-attention, as it can handle the modeling and generation of sequential data. It has advantages in handling global information, but its ability to process local information is relatively weak. The architecture of the above two image processing methods is shown in Fig. 3. By combining CNN with Transformer, local and global information in images can be effectively captured and processed. At present, researchers mainly fuse the two through methods based on serial–parallel splicing and local replacement.

Researchers have proposed many hybrid models based on the idea of structural splicing and analyzed different splicing methods. Carion et al. [39] proposed detection transformer (DETR) using a concatenation method of CNN first and then Transformer, which is the first time Transformer has been used for image object detection. This method quickly learns the global features of the input image, improving the learning speed and overall performance of the model. Dai et al. [50] proposed collaborative attention network (CoAtNet) using the same stitching strategy, which combines the advantages of CNN's translation equivalence and Transformer's input adaptive weighting, and global reception field into a single framework.

The method based on local replacement can also leverage the advantages of CNN and Transformer by replacing specific modules in the network. The current replacement methods mainly include replacing embedded blocks, replacing feedforward layers, and replacing self-attention layers. Replacing embedded blocks is a common strategy, and ViTC [40] improves the sensitivity of local spatial information and reduces dependence on positional encoding by replacing large convolution kernels with multiple small ones and embedding blocks. The way to replace the feedforward layer is to introduce deep convolution into the feedforward network and introduce locality, which can improve model performance to a certain extent. However, deep convolution deepens the network and increases model complexity. In the method of replacing self-attention layers, some scholars have introduced self-attention into CNN to mine channel contextual information and improve network performance.

However, the computational complexity of the self-attention mechanism increases squarely with the increase of context length, resulting in very low computational efficiency. Mamba [55] alleviates the modeling constraints of CNNs through global receptive fields and dynamic weighting, and provides advanced modeling capabilities similar to Transformers, as shown in Fig. 3. The core of Mamba is an improved version of the state-space model (SSM), which is a class of fully cyclic architectures used for sequence modeling with linearly increasing complexity. At the same time, Mamba adopts input-related parameterization methods to support scalable, hardware-optimized computing. With the continuous improvement of Mamba, ViM [63] converted Mamba's 1-D scanning into bidirectional 2-D cross scanning, improving the efficiency of image processing. As a grayscale image, the texture information of SAR images is manifested as the spatial correlation of pixel values. The Mamba model can effectively capture the long-range dependencies of various local regions within the image. Therefore, applying the Mamba model to SAR image tasks is expected to promote the development of artificial intelligence in the field of SAR image interpretation.

D. SAR Detection Model

There are many innovations in the architecture design of existing SAR detection networks based on deep learning, especially in SAR image noise processing, target contour feature extraction, and the combination of SAR global and local information.

Following the idea of SAR image noise processing, scholars mainly combine denoising modules with detection tasks to form an end-to-end joint optimization network that preserves ship target information and suppresses noise. Zhao et al. [56] deployed a simplified morphological denoising model in the SAR ship detection network, effectively suppressing speckle noise and enabling the network to focus on target features. Zhou et al. [57] built an SAR target detection network based on a diffusion model, introduced central pixel difference convolution to reduce the scattering intensity of background noise, and mapped the SAR detection task as a denoising diffusion process from a noise bounding box to a ground truth (GT) bounding box.

Following the idea of SAR target contour feature extraction, scholars are dedicated to design specialized convolutional layers to enhance ship edge features and improve the recognizability of ships, especially in complex background noise. Jiang et al. [58] replaced the tangent function in the original Chan–Vese model with a hyperbolic sine function to improve the accuracy of the target contour and maintain excellent computation time. Han et al. [59] considered the distortion problem caused by upsampling during network feature extraction and designed an active contour loss to learn more boundary information and improve model performance. Gao et al. [60] used deformable convolution to achieve precise fitting of the target contour and designed a contour keypoint localization method to achieve direct prediction of the target.

Following the idea of combining SAR global and local information, scholars integrate the global information of SAR images (such as background sea surface features and large-scale information of targets) with the detailed target features of local regions (such as shape, size, and texture of targets) to achieve detection tasks in complex backgrounds and weak target detection. Li et al. [61] embedded graph neural networks into detection models to extract global visual features of SAR, and used multiscale design to obtain local features, in order to achieve high-quality SAR automatic target detection. Li et al. [62] designed a global context information

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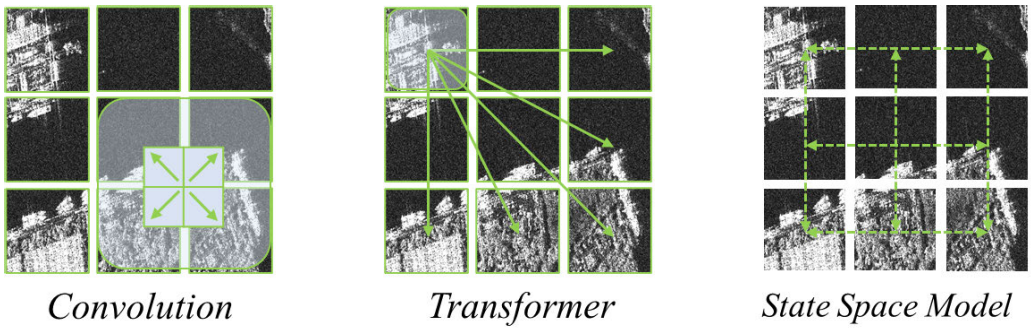


Fig. 3. Different deep learning network architectures for processing images.

Algorithm 1 GLD Block Extract SAR Ship Feature Maps

Input: Feature Maps $X = \sum_{i=1}^{B} X_i$, $X_i \in \mathbb{R}^{H\times W\times C}$
Output: Enhanced feature maps $Y = \sum_{i=1}^{B} Y_i$, $Y_i \in \mathbb{R}^{H\times W\times C}$
Global Information Augmentation Encoder (GIAE):
1: Reshaping feature maps: $X_i \in \mathbb{R}^{H\times W\times C} \rightarrow X_i \in \mathbb{R}^{HW\times C}$
X and X_{res} shape (B, H, W, C) to (B, HW, C);
2: Layer Norm processing: $X_{LN} = LN(X)$;
3: Mamba processing: $X_{out} = \text{Mamba}(X_{LN}) + X_{res}$.
Mamba's core algorithm Selective-SSM: X shape (B, L, D), with B is the batch size, L is HW and represents the length of sequence data, $Disk$ and represents the dimension size of the hidden layer, N is the state dimension;
$\bar{\boldsymbol{A}}$: (D, N), $\bar{\boldsymbol{B}}$: (B, L, N), $\bar{\boldsymbol{C}}$: (B, L, N), $\bar{\boldsymbol{D}}$: (B, L, D);
$h'(t) = \bar{\boldsymbol{A}}h(t) + \bar{\boldsymbol{B}}x(t)$, $y(t) = \bar{\boldsymbol{C}}h(t) + \bar{\boldsymbol{D}}x(t)$.
LGIM:
4: Reshape: $X_{out} \in \mathbb{R}^{HW\times C} \rightarrow X_{out} \in \mathbb{R}^{H\times W\times C}$, X_{out} shape (B, HW, C) to (B, H, W, C).
5: GIM processing: $X_{next} = \text{GIM}(X_{out})$;
6: DSA processing: $X_{dsa} = \text{DSA}(X_{next})$.
Decoder:
7: Layer Norm processing: $X_1, X_2 = LN(X_{dsa})$,
8: Feedforward neural network processing:
$Y = X_2 + \text{Linear}(\text{GeLU}(\text{DWconv}(\text{Linear}(X_1))))$.

interaction module and a local interaction attention (LIA) module based on the Transformer architecture to improve the detection accuracy of small targets in SAR images.

III. Proposed Method

In this section, we will introduce the proposed real-time SAR ship detector—GLDet. It is a one-stage end-to-end detector consisting of the backbone network, the FPN, and the detection head. Its overall structure is shown in Fig. 4.

GLDet is a supervised model that can transform detection heads to achieve different tasks (object detection, instance segmentation, and rotated object detection). Meanwhile, GLDet can achieve multitask learning for object detection and instance segmentation, and improve model generalization performance through cross training.

A. Backbone Network

The backbone network is divided into five layers, namely, the Stem layer and Stage 1–4 layers.

TABLE I
Architecture of GLDet

Network layer	Operator	Input Channel	Out Channel	Stride
Stem	7×7 Conv+BatchNorm+ReLU	3	12	2
	3×3 Conv+BatchNorm+ReLU	12	12	1
	3×3 Conv+BatchNorm+ReLU	12	12	1
	3×3 Conv+LayerNorm	12	24	1
Stage1	3×3 Conv	24	48	2
	GLD Block	48	48	-
Stage2	3×3 Conv	48	96	2
	GLD Block	96	96	-
Stage3	3×3 Conv	96	192	2
	GLD Block	192	192	-
Stage4	3×3 Conv	192	384	2
	GLD Block	384	384	-

Stem layer first uses 7 × 7 convolution to compress feature maps and expand feature channels, which extracts semantic information using the receptive field of large convolution kernels. Afterward, two 3 × 3 convolutions are used to mine local information. Finally, after expanding the channels, LayerNorm is used to enhance the stability of network training. The specific structural details are shown in Table I. The output channel C is 12.

Stage 1–4 layers consist of convolution and global information augmentation encoder–local gradient information miner–decoder (GLD) blocks. The convolution stride is 2 and is used for downsampling the feature map. The GLD block performs fine feature processing, consisting of a global information augmentation encoder (GIAE), a LGIM, and a decoder. The algorithm process is shown in Algorithm 1, and the algorithm structure is shown in Fig. 5.

1) Global Information Augmentation Encoder: GIAE reshapes input feature maps into discrete sequence data and uses the Mamba module to complete long contextual tasks. The structure of Mamba is shown in the top right corner of Fig. 5. The data are processed through a two-branch structure, with one part only undergoing linear activation and the other part undergoing selective-SSM (Selective-SSM) processing after linear activation. The two are then multiplied by a matrix to complete the long-distance modeling. Finally, the global features are enhanced by adding them to the input sequence data. The core of Mamba is Selective-SSM, which captures the dynamic changes of sequence data through hidden states.

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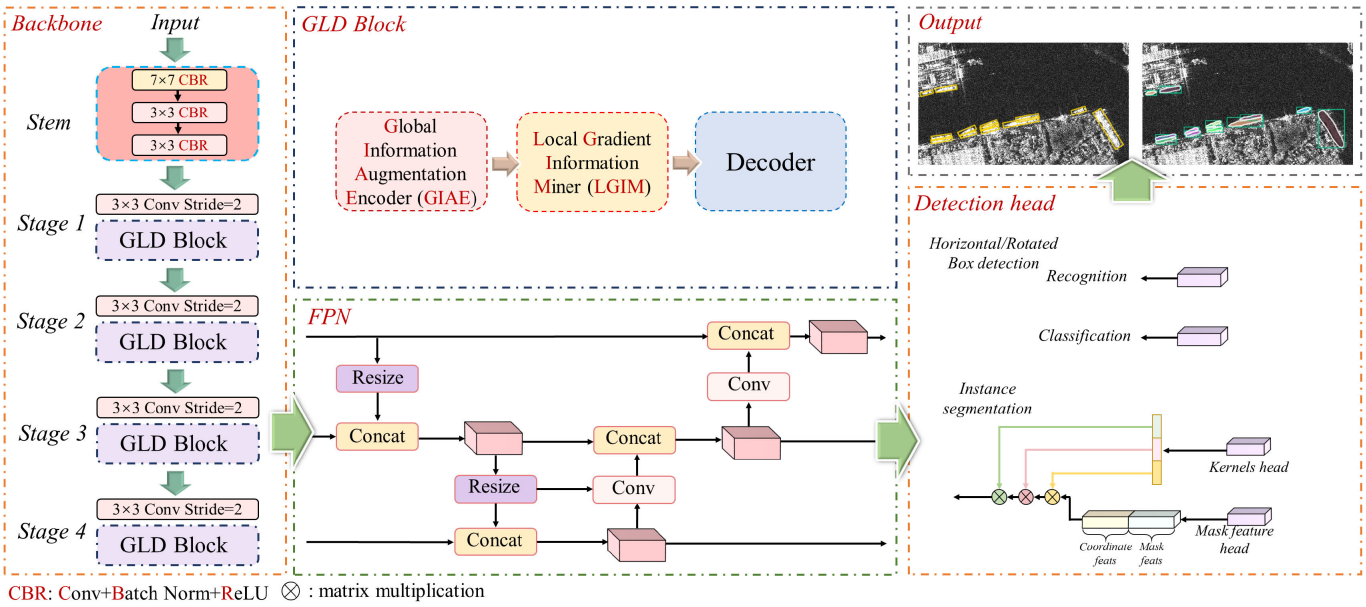


Fig. 4. Structure of the real-time SAR ship detector—GLDet.

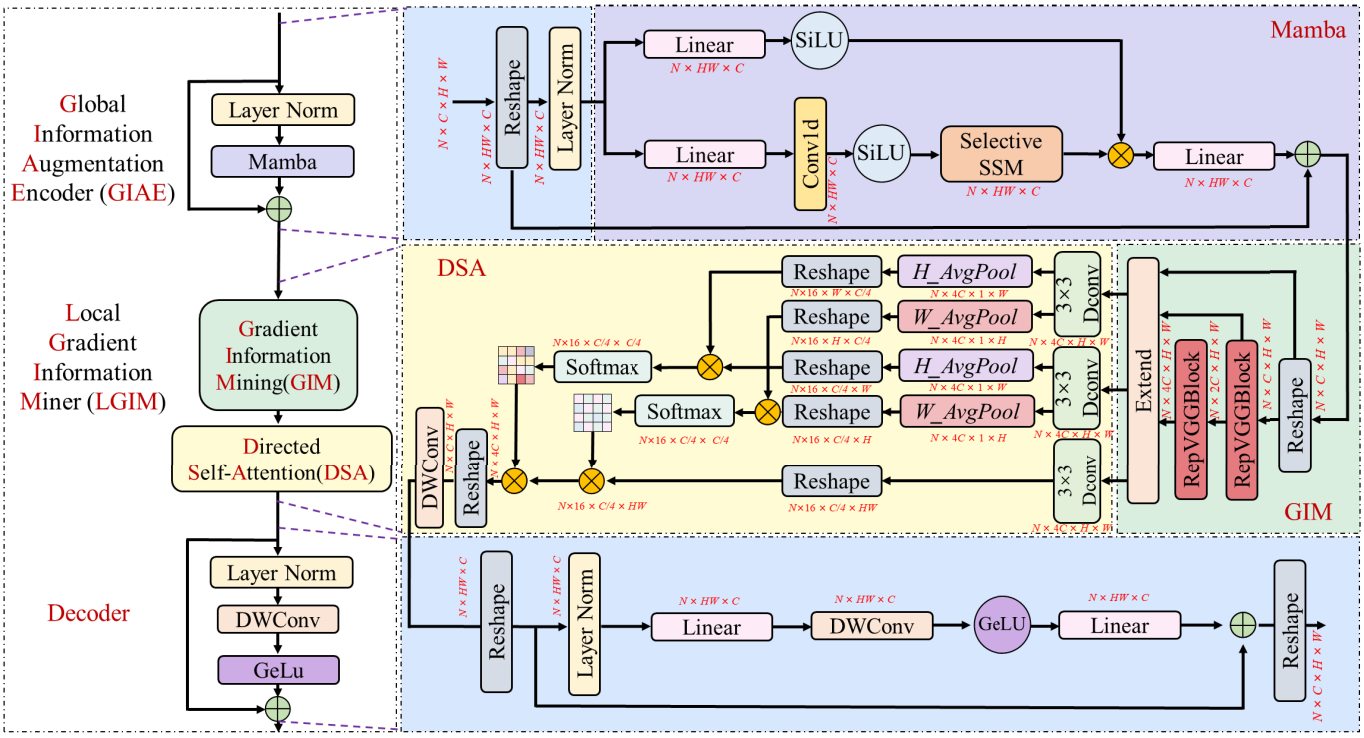


Fig. 5. Algorithm structure of GLD block.

Selective-SSM is a mathematical model used to describe and analyze the behavior of dynamic systems, capturing long-term dependencies in data by mapping sequential data to the state space. Multiplying the input with matrix $\bar{B}$ yields a vector that represents the impact of the input on the system state. State representation is an implicit vector that is multiplied by matrix $\bar{A}$ to describe the interrelationships between internal states. Update the state based on the current state and input before outputting. Matrix $\bar{C}$ transforms the state into an output, describing the relationship between the state and the output. Matrix $\bar{D}$ connects inputs and outputs. According to the formula of Selective-SSM (shown in Algorithm 1), expand the state formula

$$\begin{aligned} h_0 &= \overline{\boldsymbol{B}} x_0 \\ h_1 &= \overline{\boldsymbol{A}}\overline{\boldsymbol{B}} h_0 + \overline{\boldsymbol{B}} x_1 \\ h_2 &= \overline{\boldsymbol{A}}^2 \bar{B} h_0 + \overline{\boldsymbol{A}}\overline{\boldsymbol{B}} h_1 + \overline{\boldsymbol{B}} x_2 \\ &\cdots \end{aligned} \tag{1}$$

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And the output based on the current state is

$$\begin{aligned} y_0 &= \overline{\boldsymbol{C}}\overline{\boldsymbol{B}}x_0 \\ y_1 &= \overline{\boldsymbol{C}}\overline{\boldsymbol{A}}\overline{\boldsymbol{B}}x_0 + \overline{\boldsymbol{C}}\overline{\boldsymbol{B}}x_1 \\ y_2 &= \overline{\boldsymbol{C}}\overline{\boldsymbol{A}}^2\overline{\boldsymbol{B}}x_0 + \overline{\boldsymbol{C}}\overline{\boldsymbol{A}}\overline{\boldsymbol{B}}x_1 + \overline{\boldsymbol{C}}\overline{\boldsymbol{B}}x_2 \\ &\cdots \end{aligned} \tag{2}$$

At time k, the output y_k can be represented as

$$\begin{aligned} y_k &= \overline{\boldsymbol{C}}\overline{\boldsymbol{A}}^k\overline{\boldsymbol{B}}x_0 + \overline{\boldsymbol{C}}\overline{\boldsymbol{A}}^{k-1}\bar{B}x_1 + \overline{\boldsymbol{C}}\overline{\boldsymbol{A}}\overline{\boldsymbol{B}}x_{k-1} + \overline{\boldsymbol{C}}\overline{\boldsymbol{B}}x_k \\ \Rightarrow y &= x * \bar{K} \\ \bar{K} &= \left(\overline{\boldsymbol{C}}\overline{\boldsymbol{A}}^k\overline{\boldsymbol{B}}\right)_{k\in|L|} \\ &= \left(\overline{\boldsymbol{C}}\overline{\boldsymbol{B}}, \overline{\boldsymbol{C}}\overline{\boldsymbol{A}}\overline{\boldsymbol{B}}, \ldots, \overline{\boldsymbol{C}}\overline{\boldsymbol{A}}^k\overline{\boldsymbol{B}}, \ldots, \overline{\boldsymbol{C}}\overline{\boldsymbol{A}}^{L-1}\overline{\boldsymbol{B}}\right). \end{aligned} \tag{3}$$

From the formula, it can be seen that Selective-SSM is a linear recursive computation that uses convolution to calculate states. The core of Mamba is linear, and in order to increase the nonlinear expression ability of the model, rectified linear unit (ReLU) activation functions are added during the state update and output generation processes. SAR images contain rich reflection information, and background interference and noise impose a burden on interpretation tasks. Mamba adopts an information selection mechanism to dynamically select important state information based on assigned weights, highlighting target features. Meanwhile, its feedback mechanism can evaluate the output based on the current task requirements, adjust the parameters for feature extraction and weight allocation, and focus on the feature channels of interest to reduce the impact of interference.

The features of SAR images often include complex texture information and spatial correlation. Mamba's linear SSM is based on linearly designed state transition equations and observation equations, modeling the long-range dependence relationship between targets and backgrounds in SAR images, effectively mining the dynamic process of SAR targets, and adapting to changes in the scene where the targets are located without losing information due to complex nonlinear processing. In SAR data, many features have relatively clear structures and regularities, and linear models can provide accurate modeling in such structured data.

SAR data can be very complex in certain situations, especially when the dataset is small or noisy, traditional deep neural networks are prone to overfitting problems. Mamba's linear structure can better adapt to the true distribution of data through efficient regularization and parameter optimization, thereby effectively avoiding overfitting and improving the model's generalization ability. At the same time, Mamba has higher computational efficiency, which can reduce the consumption of computing resources while ensuring high accuracy.

In summary, Mamba effectively captures the entire image, addressing the inherent positional sensitivity of visual data and making it more suitable for HR images.

2) Local Gradient Information Miner (LGIM): LGIM consists of the GIM module and DSA.

GIM is a drip network composed of RepVGGBlocks, as shown in Fig. 6. It uses the multibranch structure (3 × 3 convolution + 1 × 1 convolution +identity mapping) to generate gradient feature maps. The magnitude of the gradient represents the contribution of each pixel in the input image or intermediate feature map to the final loss function change. A large amplitude indicates that a certain feature or region has a greater impact on the final prediction result, while a small amplitude indicates that the feature or region contributes less to the change of the loss function. The schematic of the feature gradient is shown on the right side of Fig. 6. Through gradient reflection, GLDet can focus more on local important features in the input, helping the model improve sensitivity to each local region, enhance attention to target contour information, and improve training convergence ability. GIM is responsible for generating feature maps with diverse styles, which are then sent to DSA for channel attention.

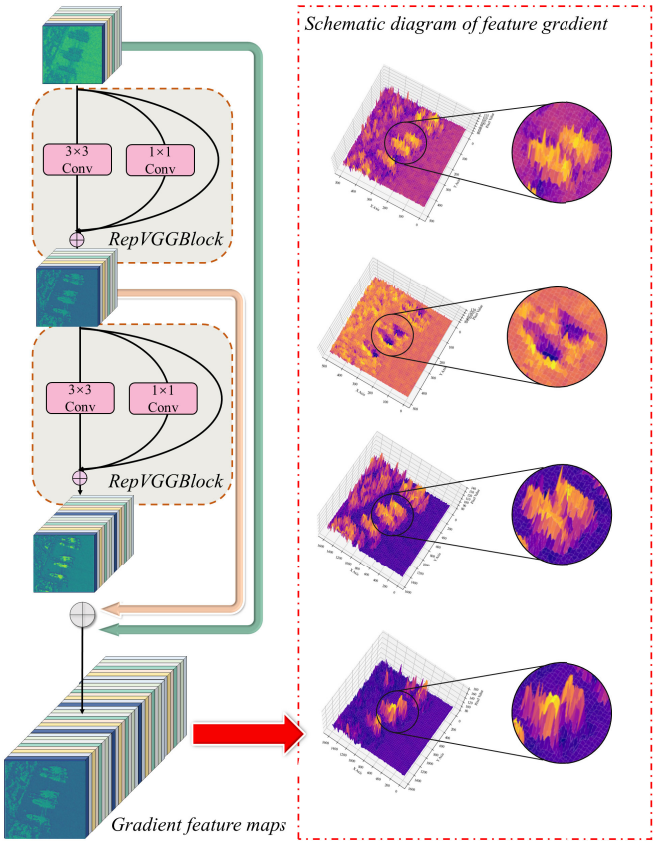


Fig. 6. Schematic of GIM.

The DSA structure is shown in the middle of Fig. 5, which is used to mine feature channel information and focus on feature channels that are more suitable for multiclass tasks while performing feature association. Features are fed into three branches for 3 × 3 depthwise convolution processing to emphasize local information. Two branches are subjected to H_AvgPool and W_AvgPool processing and feature map reshaping, followed by cross matrix multiplication to obtain the cross channel cross-covariance matrix, which is the implicit attention map that contains global contextual information. When reshaping, the feature map size changes from (N, C, H, W) to (N, Head, C/Head, HW). Among them, Head can help to reduce the computational cost of matrix multiplication by setting it to 16. The scale of the final feature map is $R^{(C/\text{Head})\times(C/\text{Head})}$. Unidirectional pooling operations can accelerate network processing, and feature maps generated

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in both directions help the network focus more comprehensively on key channels and improve detection capabilities. The process definition of DSA is

$$\begin{aligned}\mathbf{X}_1, \mathbf{X}_2, \mathbf{X}_3 &= \mathrm{Dconv}_{3\times3}(\mathbf{X})\\ \hat{\mathbf{X}} &= \mathrm{Attention}_H \cdot \mathrm{Attention}_W \cdot \mathbf{X}_3\\ \mathrm{Attention}_W &= \mathrm{Softmax}\big(\mathrm{Reshape}\big(\mathrm{Avg}_W(\mathbf{X}_1)\big)\big)\\ &\quad \cdot \big(\mathrm{Reshape}\big(\mathrm{Avg}_W(\mathbf{X}_2)\big)\big)\\ \mathrm{Attention}_H &= \mathrm{Softmax}\big(\mathrm{Reshape}\big(\mathrm{Avg}_H(\mathbf{X}_1)\big)\big)\\ &\quad \cdot \big(\mathrm{Reshape}\big(\mathrm{Avg}_H(\mathbf{X}_2)\big)\big). \end{aligned} \tag{4}$$

3) Decoder: Decoder is composed of a feedforward neural network, aimed at helping the model better understand the local and global structures in sequence data, and further improving the model's expressive power. Feedforward network (FFN) consists of two linear transformations (fully connected layers), a depthwise separable convolution, and a nonlinear activation function Gaussian error linear unit (GeLU). The GeLU function is smoother, which helps improve the convergence speed and performance of the training process. The process definition of decoder is

$$\begin{aligned}\mathbf{X}_1, \mathbf{X}_2 &= \mathrm{Reshape}(\mathbf{X})\\ \hat{\mathbf{X}}_2 &= \mathrm{Linear}(\mathrm{GeLU}(\mathrm{Dconv}(\mathrm{Linear}(\mathbf{X}_2))))\\ \mathbf{Y} &= \mathrm{Reshape}\big(\mathbf{X}_1 + \hat{\mathbf{X}}_2\big). \end{aligned} \tag{5}$$

B. Feature Pyramid Network

FPN aims to extract image features at different scales and enhance feature expression ability and information flow. We adopt an improved version of FPN, path aggregation feature pyramid network (PAFPN), as the neck of GLDet. PAFPN added a bottom–up path on top of FPN. Specifically, this enhancement initiates from the deepest feature layer and progressively merges adjacent pyramidal levels through lateral connections. At each step, the current layer's features are fused with the higher-resolution features from the preceding level through upsampling and element-wise addition operations. This hierarchical fusion propagates semantic information upward through all feature levels until reaching the topmost layer, ultimately generating enhanced multi-scale representations with both rich semantic content and precise spatial details. PAFPN processes the feature maps of the C_3, C_4, and C_5 three-layer networks, and the formula is shown as follows:

$$\begin{aligned}P_4' &= \mathrm{Upsample}(C_5) + C_4\\ P_3 &= \mathrm{Upsample}\big(P_4'\big) + C_3\\ P_4 &= \mathrm{Conv}(P_3) + P_4'\\ P_5 &= \mathrm{Conv}(P_4) + C_5. \end{aligned} \tag{6}$$

C. Detection Head

In order to train a one-stage object detector, the prediction of each target will be matched with the GT bounding box through a label allocation strategy. This article adopts a dynamic soft label allocation strategy based on SimOTA as the training strategy for object detection tasks. Meanwhile, the detection head can be fine-tuned according to the requirements of different tasks.

1) Object Detection Task: The cost function of the dynamic soft label allocation strategy is

$$C = \lambda_1 C_{\mathrm{cls}} + \lambda_2 C_{\mathrm{reg}} + \lambda_3 C_{\mathrm{center}} \tag{7}$$

where C_{cls},C_{reg}, and C_{center} correspond to the classification cost, region prior cost, and regression cost, respectively. λ_1,λ_2, and λ_3 represent the weight of three costs. λ_1 and λ_3 are set to 10, and λ_2 is set to 3.

This article uses quality focal loss (QFL) as C_{cls}. QFL directly integrates the localization quality of the target into the classification loss

$$C_{\mathrm{cls}} = -|y - P|^2((1-y)\log(1-P) + y\log(P)). \tag{8}$$

P represents the predicted score of the target. If the predicted target is determined to be a positive sample, then y is 1; otherwise, it is 0.

Generalized intersection over union (GIoU) is used as C_{reg}, which has scale invariance and can well reflect the overlap between prediction boxes and GT boxes

$$C_{\mathrm{reg}} = \mathrm{IoU} - \frac{|A - B|}{|A|} \tag{9}$$

where A represents the area of the predicted box and the GT box enclosed by a rectangular frame. B represents the union of the prediction box and the GT box. IoU represents the intersection and union ratio between the predicted box and the GT box.

C_{center} uses the soft center region cost, and the formula is as follows:

$$C_{\mathrm{center}} = \theta^{|X_{\mathrm{pred}} - X_{\mathrm{gt}}|} - \mu \tag{10}$$

where $|X_{\mathrm{pred}} - X_{\mathrm{gt}}|$ represents the center distance between the predicted box and the GT box. θ and μ are the hyperparameter and set to 10 and 3, respectively.

2) Instance Segmentation Task: Adding a branch on top of the basic framework can complete instance segmentation, as shown in Fig. 4. This branch is composed of kernel head and mask feature heads. The kernel head predicts the instance target and decomposes the prediction vector into three dynamic convolution kernels, which interact with mask features and two relative coordinate features to generate instance segmentation masks. The mask feature head predicts mask features from multiscale feature maps extracted from the neck. This branch uses diceloss as the supervised loss for instance segmentation masks

$$\mathrm{diceloss} = 1 - \frac{2 \times |P \cap G|}{|P| + |G|} \tag{11}$$

where $|P \cap G|$ represents the number of pixels where the predicted result intersects with the true label. $|P|$ and $|G|$ represent the number of pixels in the predicted result and the actual label, respectively.

3) Rotated Object Detection: Rotated object detection is similar to object detection, so only fine-tuning of the object detection head is required. Add a branch at the regression branch to predict the target angle, modify the bounding box encoder to fit the data form, and set the loss function of regression cost to Rotated_IoU_Loss.

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TABLE II
SPECIFIC SITUATION OF SAR DATASET

Dataset	Resolution(m)	Band	Polarization	Satellites	Images	Instances	Ins/Img
SSDD/SSDD+ [31]	1~15	C, X	HH, VV, VH, HV	S-1, RadarSat-2, TerraSAR-X	1,160	2,587	2.23
HRSID [51]	0.5~3	C, X	HH, HV, VH, VV	S-1B, TerraSAR-X, TanDEMX	5,604	16,969	3.03
RSDD-SAR [35]	2~20	C, X	HH, HV	GF-3, TerraSAR-X	7,000	10,263	1.47
SARDet-100K [52]	0.1~25	C, X, Ka, Ku	HH, VV, VH, HV, VV/VH, VV/HH	GF-3, TerraSAR-X, RadarSat-2, Sentinal-1	104,985	221,450	2.11

IV. EXPERIMENTS AND ANALYSIS

A. Datasets and Experimental Setup

1) Datasets: To verify the performance of GLDet, the following dataset was selected for experiments. The specific situation of the SAR dataset is shown in Table II.

a) SSDD/SSDD+ [31]: SSDD is the world's first publicly available dataset specifically designed for ship target detection in SAR images. The data are sourced from publicly available SAR images, including four polarization modes: HH, HV, VV, and VH, with a resolution of 1–15 m. A total of 1160 images and 2456 ships are included, with a relatively small collective data volume. The SSDD only annotates ships with pixel values greater than 3. SSDD+ transforms the vertical border into a rotated border, which enables direction estimation of the target while completing detection tasks.

b) HR SAR image dataset (HRSID) [51]: HRSID contains a total of 5604 HR images and 16 951 ship instances, including multiple polarization modes, sea conditions, and scenarios. The resolution of SAR images is 0.5–3 m.

c) RSSD in SAR images (RSDD-SAR) [35]: RSDD-SAR consists of 84 data from GF-3 satellite, 41 data from TerraSAR-X satellite, and two uncut large images. It includes 7000 images with multiple imaging modes, polarization modes, and resolutions, as well as 10 263 ship instances.

d) SARDet-100K [52]: SARDet-100K dataset is the first large-scale SAR target detection dataset, containing over 100 000 annotated images and a total of six types of targets, including ship, aircraft, car, tank, bridge, and harbor.

In object detection and instance detection tasks, we use SSDD for ablation experiments, and use SSDD and HRSID for comparative experiments. The dataset images of SSDD and HRSID are trained and detected at 800 × 800 pixels.

In the rotated object detection task, we used SSDD+ for ablation experiments, and use SSDD+ and RSDD-SAR for comparative experiments. The dataset images of SSDD+ and RSDD-SAR are trained and detected at 512 × 512 pixels.

We conducted generalization experiments using two large images from Terra-SAR and GF-3 satellites to verify the generalization of the instance segmentation model and the rotated object detection model. Finally, use the SARDet-100K dataset for discussion.

2) Experimental Setup: We use CUDA12.1, PyTorch2.1.2, and OpenMMLab algorithm systems as algorithm platforms. We use the standard common objects in context (COCO) format in the experiment. The experiment is conducted on an NVIDIA GTX 3090 graphics card. Using the AdamW optimizer, the basic learning rate is 0.004, the rating threshold is set to 0.01, and the batch size is set to 16. To ensure the fairness of ablation and comparative experiments, all experiments are trained for 500 epochs.

3) Evaluating Indicator: This article uses Microsoft COCO (MS COCO) evaluation indicators to quantitatively evaluate the model.

In object detection and instance segmentation tasks, use $\mathrm{IoU}_{\mathrm{Bbox}}$ and $\mathrm{IoU}_{\mathrm{Mask}}$ to evaluate model accuracy

$$\mathrm{IoU}_{\mathrm{Bbox}} = \frac{\mathrm{Bbox}_{\mathrm{pred}} \cap \mathrm{Bbox}_{\mathrm{gt}}}{\mathrm{Bbox}_{\mathrm{pred}} \cup \mathrm{Bbox}_{\mathrm{gt}}} \tag{12}$$

$$\mathrm{IoU}_{\mathrm{Mask}} = \frac{\mathrm{Mask}_{\mathrm{pred}} \cap \mathrm{Mask}_{\mathrm{gt}}}{\mathrm{Mask}_{\mathrm{pred}} \cup \mathrm{Mask}_{\mathrm{gt}}} \tag{13}$$

where $\mathrm{Bbox}_{\mathrm{pred}}$ represents the predicted bounding box, and $\mathrm{Bbox}_{\mathrm{gt}}$ represents the GT box. $\mathrm{Mask}_{\mathrm{pred}}$ represents the predicted mask, and $\mathrm{Mask}_{\mathrm{gt}}$ represents the GT mask.

Evaluate various average precision (AP) values by setting different IoU thresholds. AP is the average accuracy of ten IoU thresholds ranging from 0.5 to 0.95, with a step size of 0.05. AP^{50} is the model accuracy at an IoU threshold of 0.5, and AP^{75} is the model accuracy at an IoU threshold of 0.75. AP^{S}, AP^{M}, and AP^{L} are used to evaluate the multiscale capability of the model, corresponding to targets smaller than 32^2 pixels, larger than 32^2 pixels, smaller than 64^2 pixels, and larger than 64^2 pixels, respectively. $\mathrm{AP}_{\mathrm{Mask}}$ and $\mathrm{AP}_{\mathrm{Bbox}}$ represent the accuracy of object detection and instance segmentation, respectively.

In the rotated object detection task, use AP^{50} and Recall to evaluate model accuracy. In addition, floating point operations (FLOPs) and Parameters are used to evaluate the complexity of the model, and frames per second (FPS) is used to evaluate the real-time performance of the model.

B. Ablation Experiment

In this section, we validated the effectiveness of various components in the backbone network through intuitive observation of feature maps and ablation experiments based on object detection and instance segmentation tasks.

For intuitive visual comparison, shallow feature maps of multiple ablation experiments were extracted, as shown in Fig. 7. Targets with clear contour information and cleaner background in Fig. 7(a) are beneficial for detection and segmentation tasks. In contrast, the ability of Mamba to capture features and suppress noise has been validated, as the targets in Fig. 7(b) are obscured by noise. The feature maps in Fig. 7(c) are very similar, and the target does not stand out clearly from the background, which poses a risk of channel waste.

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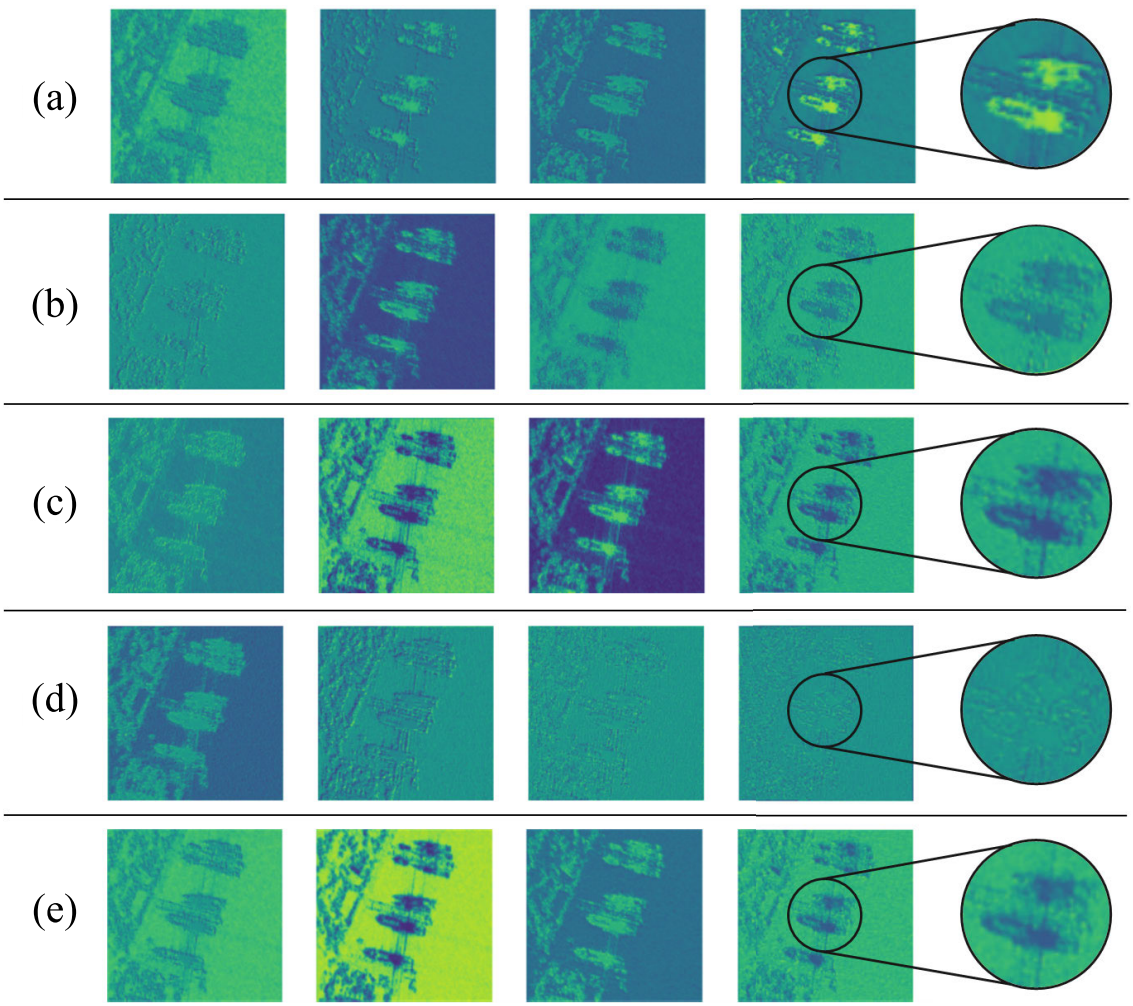


Fig. 7. Shallow feature maps of multiple networks. (a) Complete GLDet network. (b) GLDet network without Mamba module. (c) GLDet network without GIM and DSA modules. (d) GLDet network without Mamba, GIM, and DSA modules. (e) Darknet network.

Compared with Fig. 7(a), this further validates the gradient mining and feature association capabilities of GIM and DSA modules. The feature maps in Fig. 7(d) have poor quality and cannot support high-quality detection and segmentation tasks. The feature maps in Fig. 7(e) have high quality, but the target contour information is poor, which highlights the advancement of GLDet feature extraction. In summary, GLDet has achieved high-quality extraction of target features based on global semantic information enhancement and local gradient information mining.

The experimental results are shown in Table III. From the experimental results of numbers 1 and 2, it can be verified that Mamba has the ability to enhance information, achieving an improvement in detection and instance segmentation. Meanwhile, due to Mamba's own linearization design, the impact on the overall network's computational complexity and detection speed is relatively small.

From the experimental results of numbers 3–5, it can be verified that the GIM module can effectively improve the multiscale detection performance of targets. Adding the GIM module significantly improves AP^{75}_{Mask}, but its design based on depthwise separable convolutions increases the requirement for computational complexity. The combination of DSA and GIM can significantly improve detection performance, but when used alone, it has little impact on network performance. The experimental results have verified the channel context capability of GIM. In addition, mining feature maps with significant gradient information can greatly improve detection performance, but it is difficult to play a role in high-similarity channel feature maps.

From the experimental results of number 6, it can be seen that the enhancement of global information through Mamba and the mining of local information through GIM + DSA comprehensively improved the model's multiscale detection and segmentation capabilities. Each module is designed based on linear and lightweight design, allowing the model to maintain extremely fast detection speed and relatively low computational complexity.

C. Comparative Experiment

In order to verify the progressiveness of GLDet performance, which is compared with the existing advanced universal model and special SAR ship detection model.

1) Comparison With Advanced Universal Detection Models: We conducted two groups of experiments, and selected advanced general models to compare under the same experimental conditions.

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TABLE III
ABLATION EXPERIMENT BASED ON SAR INSTANCE SEGMENTATION TASK

Number	Mamba	GIM	DSA	Dataset	AP_{Mask}	AP^{50}_{Mask}	AP^{75}_{Mask}	AP_{Bbox}	AP^{50}_{Bbox}	AP^{75}_{Bbox}	FLOPs(G)	Param(M)	FPS
1	×	×	×	SSDD	65.4	94.4	80.6	66.6	95.9	80.2	20.5	22.8	**267.1**
2	✓	×	×	SSDD	67.5	95.2	82.9	67.8	96.4	83.1	21.6	22.6	255.3
3	✓	✓	×	SSDD	69.1	97.3	85.8	71.7	97.8	85.9	38.1	23.4	250.8
4	✓	×	✓	SSDD	67.9	96.0	83.9	69.2	96.9	84.9	22.6	**19.8**	250.1
5	×	✓	✓	SSDD	69.7	97.6	86.7	72.7	97.6	86.6	43.7	22.6	248.7
6	✓	✓	✓	SSDD	**71.3**	**98.2**	**89.1**	**75.7**	**98.2**	**90.0**	**45.1**	22.6	246.6

TABLE IV
COMPARISON WITH ADVANCED UNIVERSAL DETECTION MODELS (INSTANCE SEGMENTATION AND OBJECT DETECTION)

Detector	Dataset	AP_{Mask}	AP^{50}_{Mask}	AP^{75}_{Mask}	AP^{s}_{Mask}	AP^{m}_{Mask}	AP^{l}_{Mask}	AP_{Bbox}	AP^{50}_{Bbox}	AP^{75}_{Bbox}	AP^{s}_{Bbox}	AP^{m}_{Bbox}	AP^{l}_{Bbox}	FLOPs (G)	Param (M)	FPS
Mask R-CNN[13]	SSDD	58.7	89.6	73.2	57.6	61.8	28.6	62.6	91.8	76.4	63.1	64.6	20.1	115.7	44.1	12.1
Mask Scoring R-CNN [15]	SSDD	58.4	88.6	72.5	58.1	60.8	27.3	62.1	91.6	75.5	62.3	64.3	19.8	146.8	60.5	12.9
Cascade Mask R-CNN [14]	SSDD	56.5	86.8	70.3	56.3	58.9	23.6	63.2	89.8	75.1	62.3	66.1	13.1	245.5	76.8	11.1
HTC [16]	SSDD	59.5	91.8	73.2	58.9	62.1	34.6	65.7	93.8	76.5	65.1	68.6	27.6	263.2	97.1	11.7
YOLACT [17]	SSDD	49.5	88.6	53.5	18.5	53.1	42.1	55.2	90.7	60.5	55.8	48.5	12.7	**18.9**	7.9	35.5
BoxInst [21]	SSDD	45.3	92.3	82.4	62.1	70.5	50.1	68.1	92.6	86.5	70.1	79.5	40.1	209.7	34.1	25.7
RTMDet-tiny [23]	SSDD	65.7	95.9	83.0	64.4	70.5	36.9	68.2	96.0	83.3	68.3	68.8	42.9	22.2	**6.9**	**254.3**
DiffusionDet [64]	SSDD	—	—	—	—	—	—	71.9	97.2	87.5	70.3	77.6	70.6	—	—	38.9
DINO [67]	SSDD	—	—	—	—	—	—	62.6	92.2	75.2	58.9	73.6	66.7	131.1	47.5	244.2
GLDet	SSDD	**71.3**	**98.2**	**89.1**	**68.7**	**78.7**	**80.0**	**75.7**	**98.2**	**90.0**	**73.4**	**82.8**	**87.2**	45.1	22.6	246.6
Mask R-CNN [13]	HRSID	55.1	86.1	65.4	54.5	62.6	19.5	65.4	87.8	75.6	67.2	68.4	15.1	115.7	44.1	7.2
Mask Scoring R-CNN [15]	HRSID	55.1	85.2	66.2	54.5	61.7	21.2	65.3	87.7	75.6	66.6	67.3	13.5	146.8	60.5	8.3
Cascade Mask R-CNN [14]	HRSID	52.1	83.1	62.8	52.1	62.1	17.2	65.1	85.5	74.4	66.2	69.1	17.2	245.5	76.8	6.9
HTC [16]	HRSID	55.3	85.2	66.7	55.1	64.1	40.0	67.2	86.2	77.3	67.8	69.3	29.1	263.2	97.1	7.4
YOLACT [17]	HRSID	39.7	72.1	42.3	40.1	46.2	10.2	48.2	75.2	53.8	51.8	35.2	10.2	**18.9**	7.9	25.3
BoxInst [21]	HRSID	43.3	82.3	62.4	52.1	60.5	10.1	48.1	82.6	76.5	60.1	60.5	12.1	209.7	34.1	15.2
RTMDet-tiny [23]	HRSID	50.2	85.8	55.8	48.4	66.2	48.7	65.7	89.6	74.4	65.6	69.6	63.9	22.2	**6.9**	**107.4**
GLDet	HRSID	**58.6**	**90.8**	**66.0**	**55.9**	**73.9**	**66.0**	**70.8**	**93.8**	**80.9**	**70.1**	**78.8**	**75.5**	45.1	22.6	99.7

In this comparative experiment, we chose two-stage algorithms including Mask RCNN [13], Mask Scoring R-CNN [15], Cascade Mask R-CNN [14], and HTC [16], as well as one-stage algorithms including YOLACT [17], BoxInst [21], RTMDet [23], DiffusionDet [64], and DINO [65]. The experimental results are shown in Table IV. Through comparative experiments on SSDD and HRSID, it has been proven that GLDet has excellent performance and extremely fast detection speed.

We selected two complex images in the SSDD for algorithmic comparison, as shown in Fig. 8. From the ship results in the inshore dense ship scene, GLDet performs well in segmenting and detecting ships with multiscale irregular distribution. In low-resolution large-scale scenes, GLDet not only performs perfect detection on inshore ships, but also efficiently identifies dense small-scale ships in coastal ports.

2) Rotated Object Detection Experiment: The results of rotated object detection are shown in Table V. GLDet accurately identifies targets by refining and capturing target feature information, resulting in a significant increase in recall rate.

We selected some inshore and offshore scene images from the RSDD for visual comparison, as shown in Fig. 9. In inshore scenarios, ships dock along the coast and blend with onshore buildings, putting detection pressure on the algorithm. In the offshore scenario, there is a huge difference in ship scale, with large ships being dozens of times the size of small ships. Rotated FCOS, S_2A-Net, R^3Det, and RTMDet-tiny algorithms all exhibit varying degrees of missed detections, while GLDet performs well and can adapt to various complex scenarios.

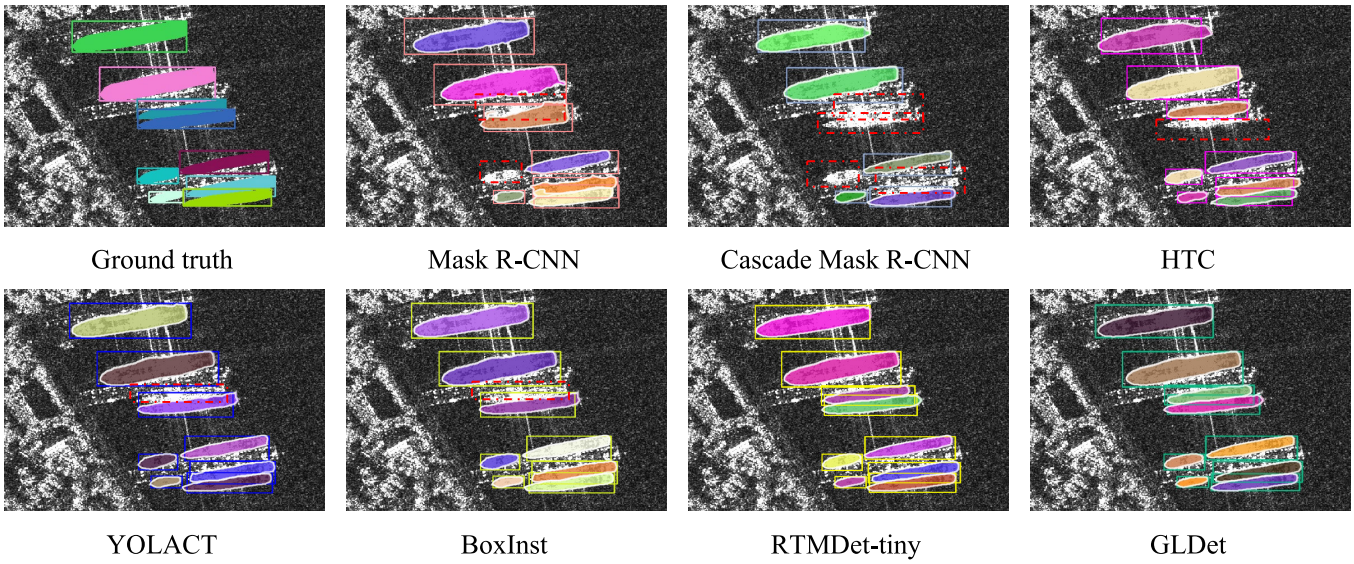
3) Comparison With Advanced SAR Ship Detection Models: In order to verify the advanced performance of GLDet, we selected some advanced SAR ship detection models for performance data comparison.

4) Instance Segmentation Experiment: The selected comparative model is as follows.

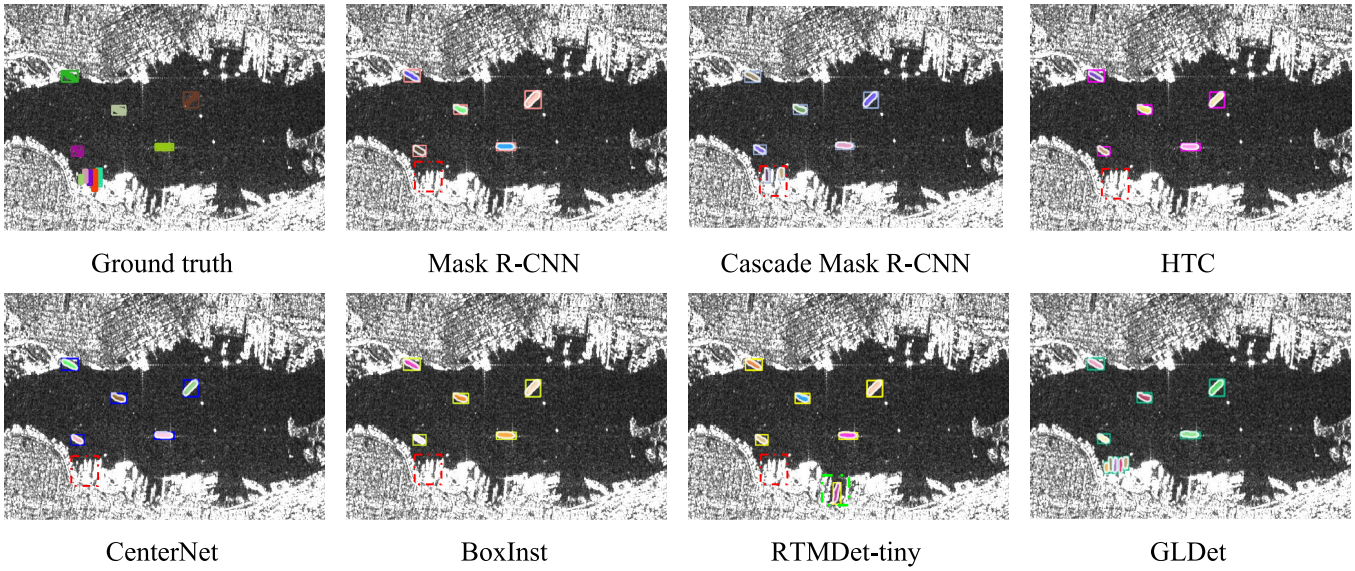
LFG-Net [41]: It is based on Mask RCNN, introduces SR denoising technology to capture image-level region and texture information, and uses HR technology to enhance Mask RoI features, achieving a certain performance improvement.

DCTC [42]: It uses discrete cosine transform (DCT) for mask prediction, improving the segmentation ability of small targets, transforming classification problems into regression problems to improve reference speed, and demonstrating strong performance on SAR images.

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(a)



(b)

Fig. 8. Comparison of the experimental results with advanced universal detection models. The red dashed box represents the missed detection target. The green dashed box represents the false detection target. (a) Dense inshore ship scene. (b) Low-resolution large-scale scene.

TABLE V
COMPARISON WITH ADVANCED UNIVERSAL DETECTION MODELS (ROTATED OBJECT DETECTION)

Detector	Dataset	AP^{50}	Recall	FLOPs(G)	Param(M)	FPS	Dataset	AP^{50}	Recall	FLOPs(G)	Param(M)	FPS
Rotated FasterRCNN [24]	SSDD+	75.1	81.6	63.4	41.1	33.5	RSDD	83.3	86.7	63.4	41.1	23.6
Oriented R-CNN [25]	SSDD+	81.0	92.3	65.6	41.3	33.0	RSDD	84.6	92.6	65.6	41.3	24.6
Rotated FCOS [26]	SSDD+	85.5	89.8	51.5	36.1	41.2	RSDD	87.5	91.2	51.5	36.1	34.2
S_2A-Net [28]	SSDD+	74.0	91.0	75.6	38.5	33.2	RSDD	78.3	90.6	75.6	38.5	42.3
R^3Det [27]	SSDD+	80.9	83.7	82.2	41.8	46.1	RSDD	86.1	89.6	82.2	41.8	46.1
RTMDet-tiny [23]	SSDD+	79.3	85.2	**10.9**	**15.3**	**201.3**	RSDD	89.4	94.2	**10.9**	**15.3**	**195.3**
GLDet	SSDD+	**90.1**	**99.1**	15.1	21.8	199.7	RSDD	**90.1**	**97.3**	15.1	21.8	186.7

RSPrompter [29]: It proposes an instance segmentation method based on the SAM and semantic category information.

SAR Ship Instance Segmentation Detector [43]: It added context fusion modules and channel attention to the HTC

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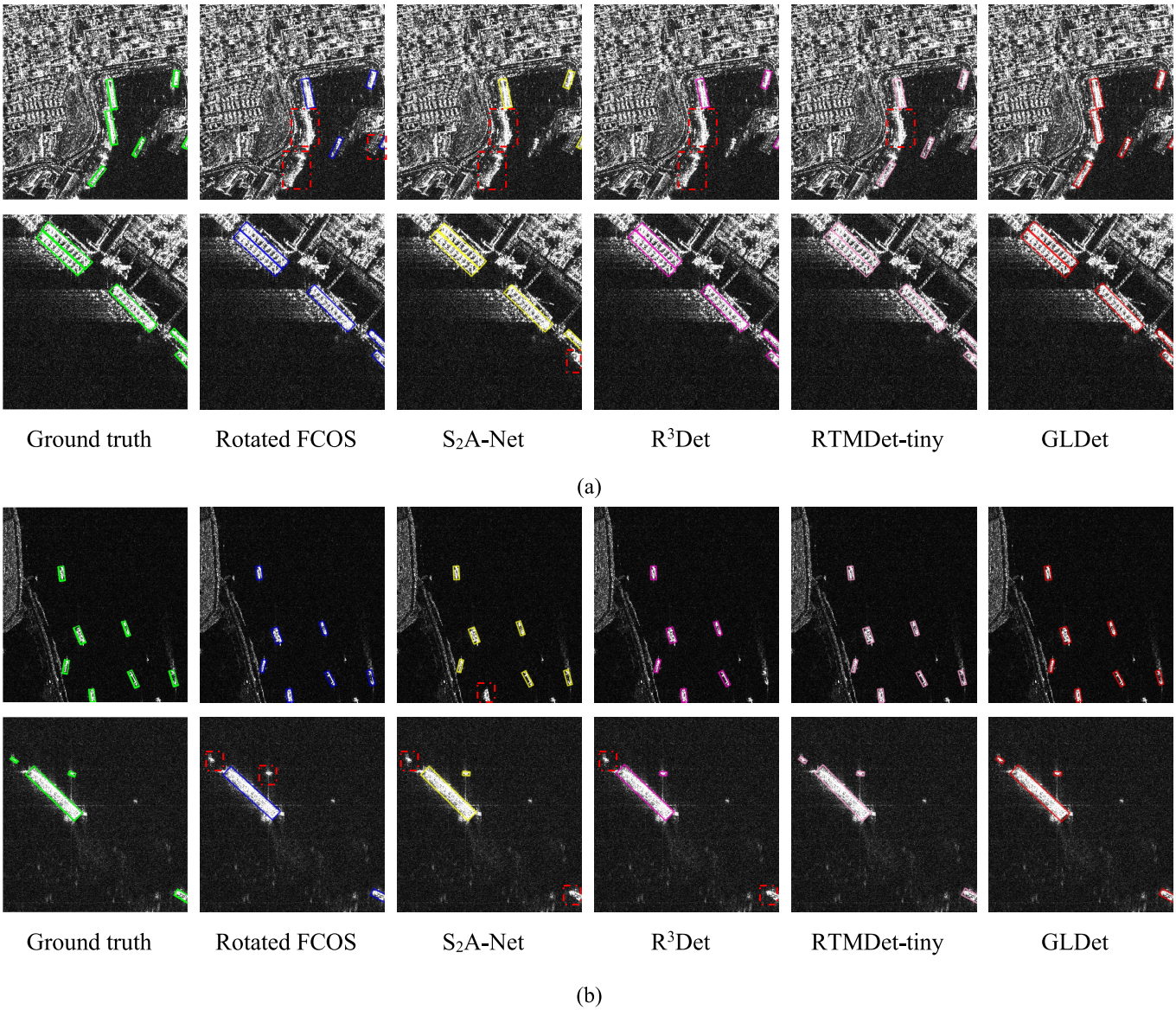


Fig. 9. Comparison of the experimental results with advanced universal detection models. The red dashed box represents the missed detection target. (a) Inshore ship scene. (b) Offshore ship scene.

model to enhance its multiscale and scene adaptation capabilities.

By comparing with the above models on SSDD and HRSID, as shown in Table VI, the comparison of instance segmentation performance of various advanced algorithms on the SSDD is shown in Fig. 10. GLDet has not only excellent segmentation ability, but also a significant speed advantage.

In order to further verify the ability of GLDet to detect both inshore and offshore scenes separately, the following algorithms were selected for comparison.

BBox-Free SAR Ship Instance Segmentation Detector [44]: It is a feature decoupling model created to address the issue of insufficient position indication information. It uses classification branches to provide guidance information for localization branches, effectively improving the algorithm's performance in ship contour segmentation.

Anchor-Free SAR Ship Instance Segmentation Detector [45]: It is a lightweight anchor-free convolutional network that dynamically transforms shared features into specific task features in the channel and spatial dimensions to enhance attention to key features. It constructs a loss function based on the centroid distance of SAR ship targets, improving the segmentation accuracy.

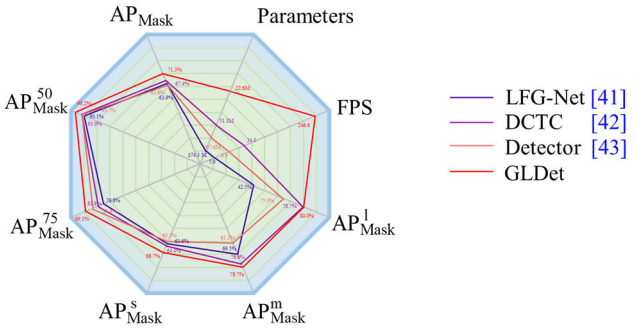


Fig. 10. Performance comparison of advanced algorithms. The more excellent each parameter is, the closer it is to the boundary. On the contrary, the worse the performance, the closer it is to the center.

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TABLE VI
COMPARISON WITH ADVANCED SAR SHIP DETECTION MODELS (INSTANCE SEGMENTATION)

Detector	Dataset	AP_{Mask}	AP^{50}_{Mask}	AP^{75}_{Mask}	AP^{s}_{Mask}	AP^{m}_{Mask}	AP^{l}_{Mask}	FLOPs(G)	Param(M)	FPS
LFG-Net [41]	SSDD	63.9	90.1	76.8	63.6	69.5	42.5	—	174.3	5.0
DCTC [42]	SSDD	67.4	93.9	83.8	64.4	76.6	**80.2**	—	51.3	34.0
RSPrompter-anchor [29]	SSDD	66.8	94.7	84.0	—	—	—	—	—	—
RSPrompter-query [29]	SSDD	67.3	95.6	84.3	—	—	—	—	—	—
Detector[43]	SSDD	62.6	93.7	78.3	63.3	61.2	75.0	—	107.6	8.9
GLDet	SSDD	**71.3**	**98.2**	**89.1**	**68.7**	**78.7**	80.0	**45.1**	**22.6**	**246.6**
DCTC [42]	HRSID	55.1	84.4	66.2	54.8	61.5	19.8	—	51.3	—
Detector[43]	HRSID	57.9	88.6	**69.5**	**57.3**	65.7	26.1	—	107.6	5.3
GLDet	HRSID	**58.6**	**90.8**	66.0	55.9	**73.9**	**66.0**	**45.1**	**22.6**	**99.7**

TABLE VII
COMPARISON WITH ADVANCED SAR SHIP DETECTION MODELS (INSTANCE SEGMENTATION)

Detector	Dataset	Scene	AP_{Mask}	AP^{50}_{Mask}	AP^{75}_{Mask}	AP^{s}_{Mask}	AP^{m}_{Mask}	AP^{l}_{Mask}
Detector [44]	SSDD	inshore	49.8	83.0	53.6	45.2	66.4	90.0
		offshore	67.1	98.7	84.3	56.6	76.4	60.0
Detector [45]	SSDD	inshore	46.0	80.6	48.3	47.9	45.7	0.0
		offshore	64.8	98.1	79.9	62.3	74.3	60.0
GLDet	SSDD	inshore	**72.2**	**98.4**	**89.3**	**66.8**	**83.8**	**100.0**
		offshore	**75.3**	**99.9**	**95.0**	**72.7**	**83.4**	**80.0**

TABLE VIII
COMPARISON WITH ADVANCED SAR SHIP DETECTION MODELS (ROTATED OBJECT DETECTION)

Detector	Dataset	AP^{50}	Recall	FLOPs(G)	Param(M)	FPS	Dataset	AP^{50}	Recall	FLOPs(G)	Param(M)	FPS
Detector [46]	SSDD+	72.2	—	31.0	**11.9**	38.1	RSDD	89.4	—	31.0	**11.9**	39.3
Detector [47]	SSDD+	**91.0**	94.5	—	42.8	19.8	RSDD	**90.8**	94.7	—	42.8	19.8
Detector [48]	SSDD+	—	—	—	—	—	RSDD	80.5	86.9	—	—	—
Detector [49]	SSDD+	87.3	89.7	—	55.8	19.7	RSDD	—	—	—	—	—
GLDet	SSDD+	90.1	**99.1**	**15.1**	21.8	**199.7**	RSDD	90.1	**97.3**	**15.1**	21.8	**186.7**

The segmentation results of different scenes are shown in Table VII, which shows that GLDet has strong scene adaptation ability and demonstrates far superior performance compared to previous advanced algorithms in both inshore and offshore scenes.

5) Rotated Object Detection Experiment: The selected comparative model is as follows.

Oriented SAR Ship Detector [46]: This model equivalently represents ships as ellipses, using a set of vectors from the center to the boundary as encoding parameters, and, based on this, achieves lightweight and efficient detection performance.

Orientation Detector for Ship Targets [47]: This model aligns deep and shallow fine-grained semantic information and designs an adaptive boundary enhancement module combined with an attention mechanism.

Oriented SAR Ship Detector [48]: This network consists of a multiscale contextual semantic information fusion module and a scattering point information learning module, which focuses on obtaining global information to improve network robustness.

Oriented Ship Detection in SAR Image [49]: This network is trained for directional ship detection based on annotations of horizontal detection boxes.

The comparative experimental results are shown in Table VIII. GLDet performs excellently on two datasets, with significantly higher recall rates than other models, and has extremely fast detection speed.

D. Generalization Experiment

In order to verify the excellent generalization ability of GLDet, we conducted a comprehensive evaluation of its performance in this experiment.

1) SAR Instance Segmentation Generalization Experiment: We first cut the GF-3 image into slices of 800 × 800 pixels. Afterward, the weights trained on the SSDD are used to detect the slices. Finally, the detected slices are merged.

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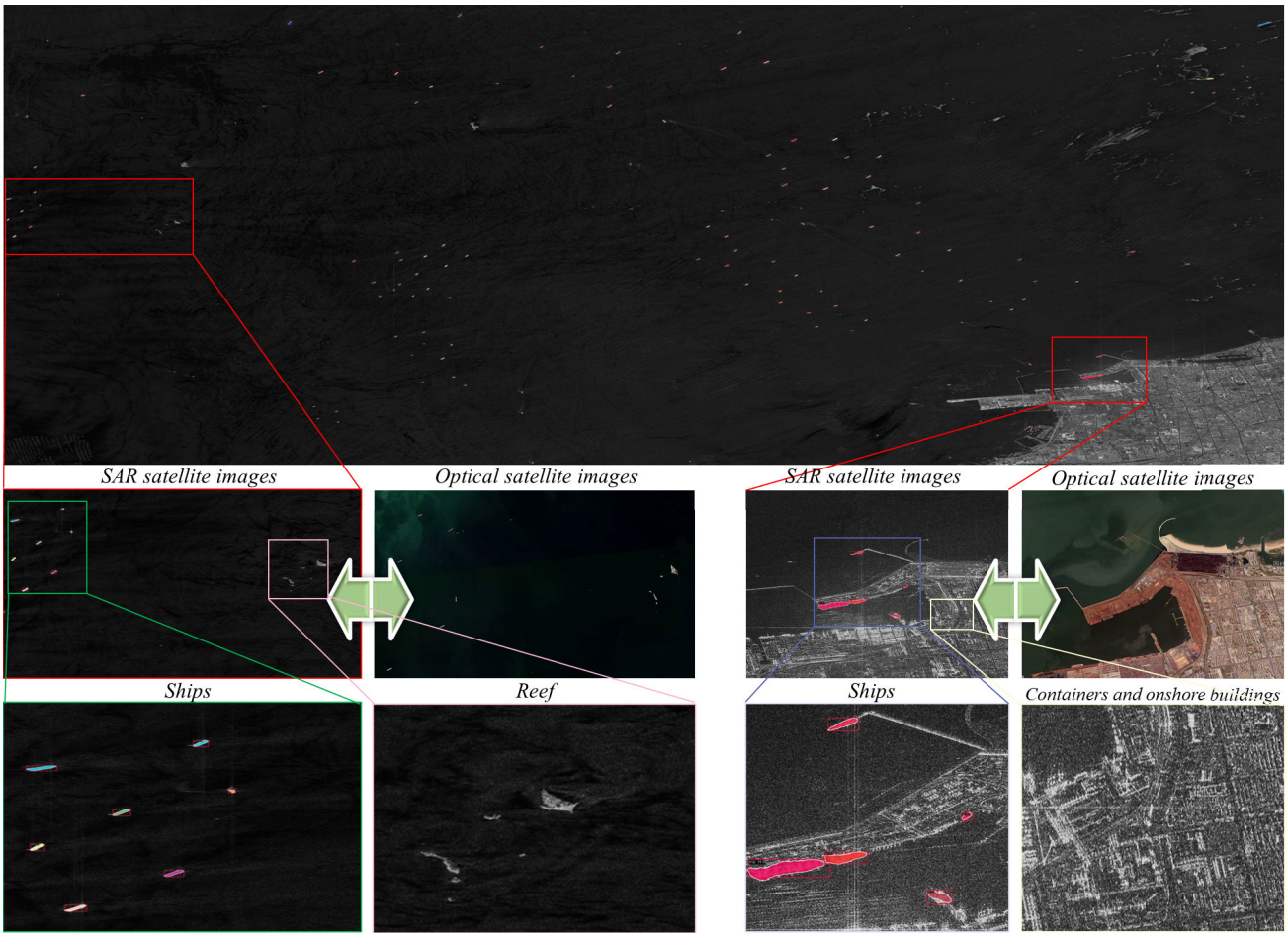


Fig. 11. Generalization experiments on GF-3 SAR satellite images.

Select the inshore and offshore areas from the large picture for detailed observation, as shown in Fig. 11. There are many buildings similar to ships (such as containers, docks, and buildings) in the inshore area. GLDet, with its outstanding global information extraction capabilities, can effectively identify ship information and further validate the effectiveness of the model in nearshore scenarios through the assistance of optical satellites for comparative observation. In the offshore area, there are reefs similar to ships. GLDet distinguishes similar features based on its ability to mine local information, effectively identifies ships, and verifies its generalization ability in offshore scenarios.

2) SAR Rotated Object Detection Generalization Experiment: To verify the performance of rotated object detection, we used the same operation as the previous experiment to segment the Terra-SAR satellite images and used weights trained on RSDD-SAR for detection. The final merged image is shown in Fig. 12.

It can be seen that GLDet once again demonstrates excellent detection capabilities, capable of distinguishing ships from complex backgrounds, and accurately detecting ships at multiple scales and angles.

V. Discussion

The rapid advancement of SAR satellite technology has driven revolutionary evolution in data interpretation, fueled by the exponential growth of HR SAR datasets. From early moderate-resolution MSAR datasets to current multimodal SAR datasets, the expansion in data volume has been paralleled by refined target classification systems. The recently released SARDet-100K dataset exemplifies this progress, encompassing 116 598 images and 245 653 annotated instances, with a breakthrough six-category taxonomy (aircraft, ships, vehicles, bridges, tanks, and harbors) that establishes new benchmarks for target recognition in complex scenarios.

Methodologically, the groundbreaking SARATR-X framework [67] has redefined the paradigm for SAR target recognition. This framework establishes a unified foundation model for multiclass SAR detection through two key innovations: 1) a multisource pretraining strategy across 14 heterogeneous SAR datasets (0.1–3-m resolution) that achieves exceptional cross-domain generalization and 2) a self-supervised contrastive learning mechanism that effectively decouples target scattering features from background clutter. While SARATR-X demonstrates outstanding performance in open-scenario detection, our proposed GLDet model complements its capabilities through a specialized contour-aware detection mechanism, showcasing superior adaptability in scenario-specific applications.

This study focuses on the interpretation of SAR ship targets and develops a detection framework that synergis-

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Fig. 12. Generalization experiments on Terra-SAR satellite images.

tically enhances global and local features. As shown in Fig. 13, when extending GLDet to full category detection on SARDet-100K, the confusion matrix reveals that GLDet achieves higher accuracy for geometrically different targets (such as ships and aircraft) through the inductive bias of specific tasks. SARATR-X performs well in generalized feature learning across different target categories, such as tanks and bridges, through large-scale pretraining. This provides a promising direction for hybrid architectures—for example, initializing GLDet's backbone network with SARATR-X pretrained weights can synergistically enhance detection capabilities.

GLDet's superior performance stems from its unique contour extraction mechanism. As shown in Fig. 14, targets like ships and aircraft exhibit distinct geometric structures that contrast sharply with backgrounds, aligning perfectly with the model's design strengths. However, for targets with weaker contour features (e.g., tanks and harbors), dielectric property differences from backgrounds often prove more critical. This highlights the fundamental distinction between the two approaches: GLDet relies on explicit geometric modeling, whereas SARATR-X employs a self-supervised scattering component to analyse implicit cluster dielectric properties.

Based on these findings, combining the dielectric feature extraction of SARATR-X with the geometric modeling of GLDet is expected to break through the current technological bottleneck. These explorations are not only expected to improve the robustness of target recognition in complex battlefield environments, but also to establish a new paradigm for the engineering deployment of SAR intelligent interpretation systems.

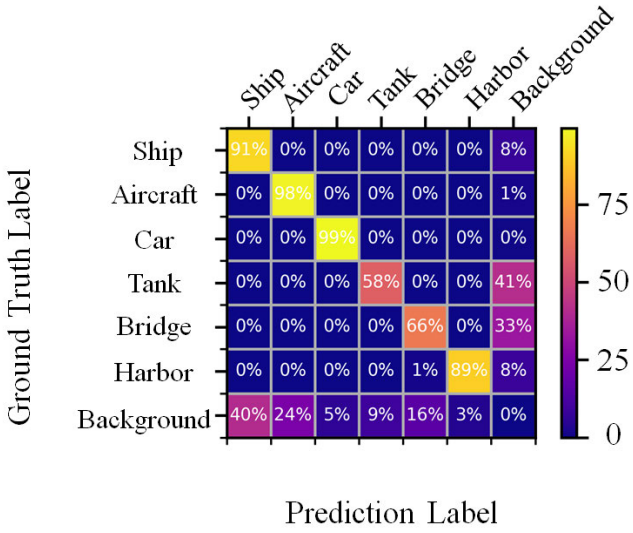


Fig. 13. Confusion matrix for SARDet-100K dataset.

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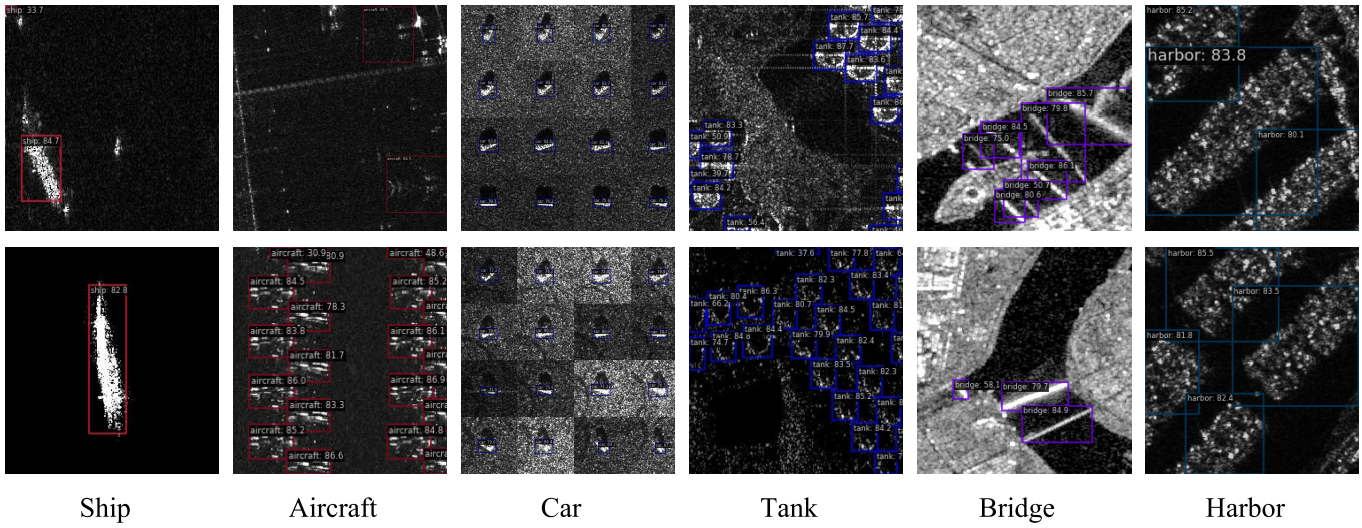


Fig. 14. SAR images of ship, aircraft, car, tank, bridge, and harbor.

VI. Conclusion

We have developed the GLDet model for SAR ship detection from a lightweight and real-time perspective. It detects ships based on global semantic information enhancement and local gradient information mining, by mining fine-grained information between classes. At the same time, GLDet has powerful multitasking detection capabilities and can complete object detection, instance segmentation, and rotated object detection tasks by changing the detection head. Excellent performance and extremely fast detection speed were achieved on the SAR ship instance segmentation datasets SSDD and HRSID, as well as the SAR ship rotated box datasets RSDD-SAR and SSDD+, verifying the robustness of the model. The good scene adaptability of GLDet was demonstrated on SAR images obtained from GF-3 and Terra-SAR satellites, validating the model's generalization ability. Finally, the model's detection ability and universality for SAR targets with significant contour information were fully validated through experiments on the large-scale dataset SARDet-100K. At the same time, it triggers thinking and guides us to continue seeking more effective target enhancement algorithms and information mining algorithms, in order to build a more comprehensive SAR detector.

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